IEORE4004: Optimization Models and Methods - Prof. Yuri Faenza 09/29/2025

Assignment 2

Deadline: 10/10/2025, h: 11:59pm

- This assignment sheet has 6 exercises. Submit your solution via Gradescope. See the syllabus for the late submission policy.
- You are allowed to discuss the assignment with others but the write-up must be individual work. Please mention in your write-up all the people you have discussed the solution with.

a) LP modelling

Problem 1: Suppose you want to solve the following LP:

$$\max c^T x : \ Ax \leq b, \ x \geq 0$$

but unfortunately it is infeasible (think e.g. about an inventory problem where the demand cannot be satisfied by the warehouse). Let $A \in \mathbb{R}^{m \times n}$. Now suppose you can "augment" your LP by "buying" some more right-hand side in your problem (think e.g. of buying some of the product from other warehouses). In particular, for $i = 1, \ldots, m$, if you want to increase the right-hand side of the i-th constraint by some value λ_i, you will pay $d_i \lambda_i$ for some fixed number d_i. Suppose moreover that the right-hand side of the i-th constraint can be augmented by at most k_i, for $i = 1, \ldots, m$. How can you find the optimal augmentation, i.e. the one that maximizes the profit of the optimal solution of the augmented LP minus the cost for the augmentation?

Problem 2: Consider the following feasible region:

$$P = \{x : Ax = b, \ x \geq 0\}.$$

Now suppose that you have two objective functions: a "primary" $c^T x$ and a "secondary" $d^T x$ (think of the primary objective function as the main goal of the company, say profit, and of the secondary one as some organization-related issue, say a measure of how fairly the tasks are divided among the employees). Describe an algorithm that finds, among all vectors that maximize the primary objective function, the one that achieves the maximum value of the secondary objective function.

b) Flows

Problem 3: Consider a Minimum Cost Flow instance without capacities and demands / offer d_v for $v \in V$. Assume moreover that the network is complete, that is, for each pairs of nodes $u \neq v$ there is an arc (u, v) and an arc (v, u). Show that $\sum_{v \in V} d_v = 0$ is a necessary and sufficient condition for the problem to have a solution. Does the same statement hold if the problem has capacities?

Problem 4: Consider the general version of the logistic problem seen in class: we produce one good and have to serve customers in T times, where the demand of customers at time $t = 1, 2, \ldots, T$ is $d_t \geq 0$. For $t = 1, \ldots, T$, we have an upper bound of $u_t \geq 0$ on the amount of product we can produce, and the unitary production cost is $c_t \geq 0$. For $t = 1, 2, \ldots, t - 1$, we can store between time t and time $t + 1$ at most $q_t \geq 0$ units of our good, with a per-unit inventory cost of $d_t \geq 0$.

a) Formulate as an LP the problem of serving all demands at minimum cost;

b) Generate 10 random <u>feasible</u> instances with $t = 50$ and solve them using your favorite software. What do you notice about the optimal solutions?

c) Duality

Problem 5: Consider the following LP

$$
\begin{array}{llllcl}
\max & 2x_1 & -x_2 & +x_3 & & \\
\text{s.t.} & & & & & \\
& 3x_1 & +x_2 & +x_3 & \geq & 40 \\
& x_1 & -x_2 & +2x_3 & = & 15 \\
& x_1 & +x_2 & -x_3 & \leq & 30 \\
& x_1 & & & \geq & 0 \\
& x_2 & & & \leq & 0
\end{array}
$$

- Write the dual of the previous problem.
- Find (however you want – but say how in your solutions) optimal solutions to the primal and dual, or discuss why they do not exist.

Problem 6: Write the dual of the min-cost flow problem with 5 nodes seen in Section 2.1 of the Lecture notes on flow problems. Then write the dual of a generic min-cost flow problem.